Homework 11

Week 13

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1 Asynchronous Circuit Testing

Problem 1. Show and explain a ***Late detection stuck-at-fault*** in Dual-rail asynchronous One-bit Full Adder Circuit below.

Input bit for number A	Input bit for number B	Carry bit input C_{IN}	Sum bit output S	Carry bit output C_{OUT}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

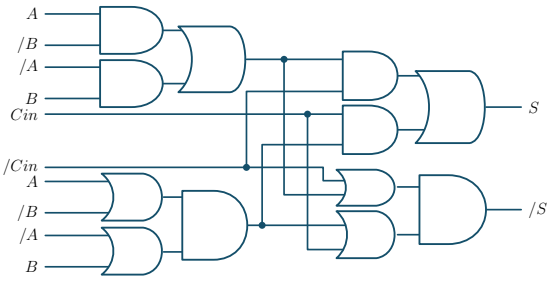


Table 1: Truth table of a full adder

Solution

Circuit equations (X = upper OR, Y = lower-left AND, P_1, P_2 = upper-right ANDs):

$$X = A\,\overline{B} + \overline{A}B, \quad Y = (A + \overline{B})(\overline{A} + B), \quad P_1 = X \cdot \overline{C_{in}}, \quad P_2 = Y \cdot C_{in},$$

$$S = P_1 + P_2, \qquad \overline{S} = (\overline{C_{in}} + X)(Y + C_{in}).$$

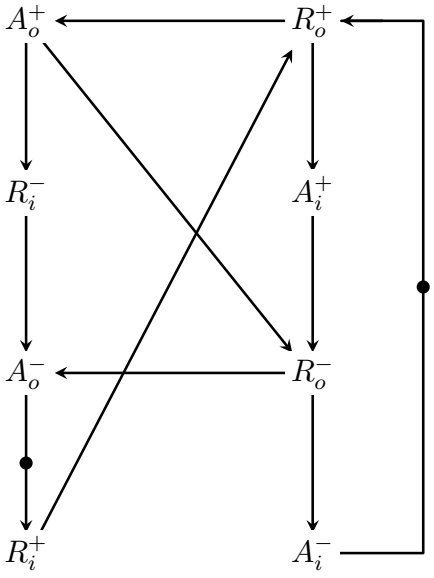
In the spacer (all rails = 0) every internal line and both outputs are 0.

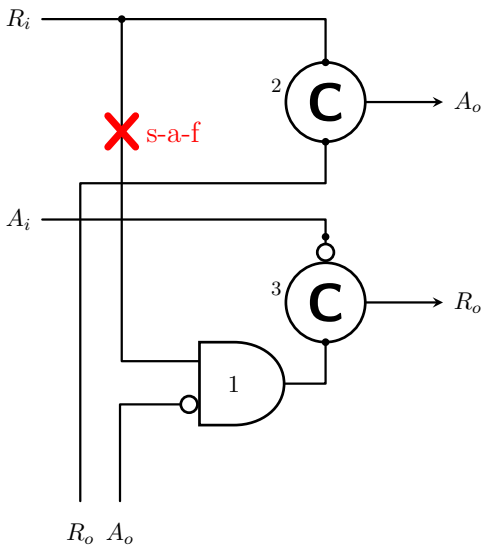
Fault: stuck-at-1 on line P_1 (output of the upper-right AND gate 1, entering the final OR gate), so $S = 1$ always.

Phase	A	B	C_{in}	P_1	S	$\overline{S}$	S_{f}	$\overline{S}_{\text{f}}$	Result
Data	1	0	0	1	1	0	1	0	correct (fault masked)
Spacer	–	–	–	0	0	0	**1**	0	no return-to-zero
Data	0	0	0	0	0	1	**1**	1	invalid code 11

For the data $ABC_{in} = 100$ the correct value of P_1 is already 1, so the output $S\overline{S} = 10$ is correct and the fault is *not* seen when the data is evaluated. It shows up only *later*: in the following spacer S cannot fall to 0, so the completion detector never sees the spacer and the handshake halts (or, for a next data word with $S = 0$, the output becomes the illegal code $S\overline{S} = 11$). Hence the fault is a **late detection** stuck-at fault.

Problem 2. Assume the control circuit below is in initial state, please describe whether the ***Stuck-at-0*** fault and ***Stuck-at-1*** fault (s-a-f) shown in the circuit lead to an **Inhibited transition** or a **Premature transition**.





Solution

Circuit equations (C = Muller C-element):

$$A_o = C(R_i,\ R_o), \qquad R_o = C\big(\overline{A_i},\ R_i \cdot \overline{A_o}\big).$$

Initial state: $R_i = A_i = A_o = R_o = 0$, so the AND gate output is 0 and the next expected transition is $R_i^+ \to R_o^+$. The fault is on the R_i input of the AND gate (1).

Stuck-at-0 ⇒ Inhibited transition. The AND output is always 0, so C-element 3 can never set $R_o = 1$. After R_i^+, the transition R_o^+ never fires, so A_o^+ (needs $R_o = 1$) never fires either and the circuit deadlocks.

Stuck-at-1 ⇒ Premature transition. The AND output becomes $\overline{A_o} = 1$. Together with $\overline{A_i} = 1$, both inputs of C-element 3 are 1 in the initial state, so R_o^+ fires immediately *before* R_i^+ arrives, violating the order $R_i^+ \to R_o^+$ of the STG.